KAIZEN MATHS

Lesson starter

Complex Numbers

Cartesian Form

Name _____

S1

Simplify each expression into the form a + bi.

Calculate (-6 - 3i) - (-5 - 5i).

S2

Expand and simplify each product into the form a + bi.

Expand and simplify (3i)(2 + 2i).

S3

Write each quotient in the form a + bi.

Write (3 + 2i) / (-1 - 4i) in the form a + bi.

S4

Find the requested conjugate and modulus.

For z = 2 - i, find z^* and $|z|$.

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ALG-D7A3-L1 · S

Show your working

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Exit ticket

Complex Numbers

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Calculate (-8 - 4i) - (4 + 8i).

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ALG-D7A3-L1 · E

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ALG-D7A3-L1 · E

Show your working

Teacher answers & working

Complex Numbers · Cartesian Form

ALG-D7A3-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Simplify each expression into the form a + bi.

Calculate (-6 - 3i) - (-5 - 5i).

Answer:

-1 + 2i

Working:

Group the real parts and imaginary parts separately.

(-6 - 3i) - (-5 - 5i) = (-6 - -5) + (-3 - -5)i.

Real part = -1.

Imaginary part = 2.

Therefore the answer is -1 + 2i.

S2

Expand and simplify each product into the form a + bi.

Expand and simplify (3i)(2 + 2i).

Answer:

-6 + 6i

Working:

Multiply every term in the first bracket by every term in the second bracket.

$(3i)(2 + 2i) = 0 + 0i + 6i + 6i^2$.

Use $i^2 = -1$, so $6i^2 = -6$.

Real part = 0 - 6 = -6.

Imaginary part = 0 + 6 = 6.

Therefore (3i)(2 + 2i) = -6 + 6i.

Teacher answers & working

Complex Numbers · Cartesian Form

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S3

Write each quotient in the form a + bi.

Write (3 + 2i) / (-1 - 4i) in the form a + bi.

Answer:

$-\frac{11}{17} + \frac{10}{17}i$

Working:

Multiply numerator and denominator by the conjugate of the denominator.

The conjugate of -1 - 4i is -1 + 4i.

$\frac{3+2i}{-1-4i} = \frac{(3+2i)(-1+4i)}{(-1-4i)(-1+4i)}$.

Denominator = $(-1)^2 + (-4)^2 = 17$.

Numerator = -11 + 10i.

So the quotient is $\frac{-11+10i}{17} = -\frac{11}{17} + \frac{10}{17}i$.

S4

Find the requested conjugate and modulus.

For z = 2 - i, find z^* and $|z|$.

Answer:

$z^* = 2 + i$, $|z| = \sqrt{5}$

Working:

The conjugate changes the sign of the imaginary part.

$z^* = 2 + i$.

Use $|z| = \sqrt{(a^2 + b^2)}$.

$|z| = \sqrt{(2^2 + (-1)^2)} = \sqrt{5}$.

Therefore $|z| = \sqrt{5}$.

E1

Simplify each expression into the form a + bi.

Calculate (-8 - 4i) - (4 + 8i).

Answer:

-12 - 12i

Working:

Group the real parts and imaginary parts separately.

$(-8 - 4i) - (4 + 8i) = (-8 - 4) + (-4 - 8)i$.

Real part = -12.

Imaginary part = -12.

Therefore the answer is -12 - 12i.

Teacher answers & working

Complex Numbers · Cartesian Form

ALG-D7A3-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E2

Expand and simplify each product into the form a + bi.

Expand and simplify (3 + i)(1 - i).

Answer:

4 - 2i

Working:

Multiply every term in the first bracket by every term in the second bracket.

$(3 + i)(1 - i) = 3 - 3i + i - i^2$.

Use $i^2 = -1$, so $-i^2 = 1$.

Real part = 3 + 1 = 4.

Imaginary part = -3 + 1 = -2.

Therefore (3 + i)(1 - i) = 4 - 2i.

E3

Write each quotient in the form a + bi.

Write (2 - 2i) / (-3 - 5i) in the form a + bi.

Answer:

$\frac{2}{17} + \frac{8}{17}i$

Working:

Multiply numerator and denominator by the conjugate of the denominator.

The conjugate of -3 - 5i is -3 + 5i.

$\frac{2-2i}{-3-5i} = \frac{(2-2i)(-3+5i)}{(-3-5i)(-3+5i)}$.

Denominator = $(-3)^2 + (-5)^2 = 34$.

Numerator = 4 + 16i.

So the quotient is $\frac{4+16i}{34} = \frac{2}{17} + \frac{8}{17}i$.